

scripts/snapshot-coverage-ledger.sh — User Guide

Last verified: 2026-08-20 (LVA-019 closure, docs/superpowers/plans/2026-08-20-workable-items-backlog-closure.md Task 4) **Inheritance:** HelixConstitution §11.4.25 (Full-Automation-Coverage Mandate) + §11.4.18 (script docs)

Overview

Freezes a per-release copy of docs/coverage-ledger.yaml under .lava-ci-evidence/coverage-ledger-snapshots/, mirroring the existing .lava-ci-evidence/distribute-changelog/<channel>/cycle-coverage-map-<version>-<code>.yaml per-release snapshot convention already used by scripts/firebase-distribute.sh for the \$6.AK device-coverage gate. This closes the "missing per-release ledgers" half of LVA-019 — the live ledger classifier itself (scripts/check-coverage-ledger.sh --strict) already had no defect; what was missing was a historical record of the ledger's state at each release moment.

What it does

1. Regenerates docs/coverage-ledger.yaml in place via scripts/generate-coverage-ledger.sh --quiet, so the snapshot reflects the current tree, not a stale committed copy.
2. Copies the freshly regenerated ledger to .lava-ci-evidence/coverage-ledger-snapshots/<version>-<code>.yaml.
3. Restores docs/coverage-ledger.yaml to its pre-run committed state (via git checkout) if the regeneration step is what dirtied it, so the script never leaves an unwanted tracked-file diff in the working tree as a side effect of taking a snapshot.

Usage

```
scripts/snapshot-coverage-ledger.sh <version>-<code>
```

Example:

```
scripts/snapshot-coverage-ledger.sh 1.3.17-1085
# -> writes .lava-ci-evidence/coverage-ledger-snapshots/
      1.3.17-1085.yaml
```

Where it's invoked from

scripts/firebase-distribute.sh, immediately after the \$6.AK Gate 7 cycle-coverage block (step "1d. LVA-019 — per-release coverage-ledger snapshot"), once per distribute run:

```
bash "$SCRIPT_DIR/snapshot-coverage-ledger.sh" "$APP_VERSION-
      $APP_VERSION_CODE" || \
echo
      "    WARNING: coverage-ledger snapshot failed (non-fatal,
      distribute continues)"
```

The call is **advisory / non-blocking** — a snapshot failure logs a warning but does not abort the distribute. The coverage ledger's own STRICT gate (scripts/check-coverage-ledger.sh --strict) already runs elsewhere in the pipeline as the actual release gate; this script is a historical-record snapshot, not a gate.

Output

- .lava-ci-evidence/coverage-ledger-snapshots/<version>-<code>.yaml — a byte-for-byte copy of docs/coverage-ledger.yaml at the moment of the run.
- docs/coverage-ledger.yaml — left unchanged in the working tree (regenerated internally to produce fresh snapshot content, then restored if it was clean beforehand).

Companion files

- scripts/generate-coverage-ledger.sh — the generator this script calls to produce fresh ledger content before snapshotting.
- scripts/check-coverage-ledger.sh — the live STRICT/advisory verifier; see docs/scripts/check-coverage-ledger.sh.md.

- docs/coverage-ledger.yaml — the committed canonical ledger this script snapshots.
- scripts/firebase-distribute.sh — the sole caller.

Cross-references

- HelixConstitution Constitution.md §11.4.25 (the mandate)
- docs/superpowers/plans/2026-08-20-workable-items-backlog-closure.md Task 4
- .lava-ci-evidence/workable-items-closures/2026-08-20-lva-019-closure.md (LVA-019 closure evidence)
- Lava CLAUDE.md §6.J/§6.L (anti-bluff posture: a snapshot mechanism that silently dirties tracked state on every run is itself the kind of side effect the Anti-Bluff Pact exists to catch)

Inheritance + classification

- Inheritance: HelixConstitution §11.4.25 + §11.4.18 + §6.J/§6.L
- Classification: project-specific (the Lava distribute-pipeline wiring is project-specific; the §11.4.25 per-release snapshot discipline is universal per HelixConstitution)

Snapshot name format guard (added 2026-08-27)

The first positional argument is now validated against a `<version>-<code>` shape before it is used as a filename.

Why

The script previously used only `${1:?...}`, which checks that the argument is non-empty and nothing more — so **any** string became a filename. A stray invocation carrying a flag produced a tracked file literally named `.lava-ci-evidence/coverage-ledger-snapshots/--check.yaml`, which was then committed. It was a verbatim copy of the ledger whose only difference was the inflated submodule count described in the generator's companion doc, it was referenced by no tracked file, and it was the *only* file in that directory — no legitimate snapshot had ever been taken there.

The guard rejects `--check` (exit 2), an empty argument (exit 1) and traversal forms such as `../evil`; it accepts real identities like `1.3.17-1085`, `0.2.11-22` and `1.2.3-1023`. A rejected invocation writes no file at all.

Known residual, deliberately not changed here: this script regenerates the tracked `docs/coverage-ledger.yaml` as a side effect and restores it only when it was clean beforehand. If the ledger is dirty mid-edit, in-progress work is silently overwritten. Changing that is a design decision, not a bug fix, so it is recorded rather than acted on.